

ANANYA SRIDHAR

Robotics & Computer Vision Engineer — Autonomous Systems, Simulation & Edge CV

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PROFESSIONAL SUMMARY

Robotics and Computer Vision Engineer with specialized expertise in physics-backed simulation (MuJoCo), multimodal sensor fusion (Extended Information Filter), Deep Reinforcement Learning, and real-time edge perception (OpenCV, DINOv2). Proven track record designing autonomous navigation agents, sub-50ms CPU vision pipelines, and interactive 3D robot kinematics simulators.

TECHNICAL SKILLS

- **Robotics & Simulation:** MuJoCo 3D Physics, Gymnasium, Sensor Fusion (EIF/EKF), Lidar SLAM / Motion Planning, Forward & Inverse Kinematics, Three.js, React Three Fiber
- **Computer Vision:** OpenCV, MediaPipe Tasks, DINOv2 (Vision Transformers), Optimal Transport (POT), Background Subtraction, Multi-Object Centroid Tracking, Edge Detection
- **AI & Deep Learning:** PyTorch, Deep Reinforcement Learning (PPO, Q-Learning), MCTS, A* Pathfinding, Scikit-Learn
- **Systems & Backend:** Python, C++, C, FastAPI, Uvicorn, WebSockets, Docker, Git, Linux, ROS Concepts, React, Vite

PROFESSIONAL EXPERIENCE

Manas Learning — AI/ML Engineer Intern (Perception & Systems)

Mar 2026 – Present | Remote / Mumbai, India

- **Real-Time Visual Biofeedback:** Integrated MediaPipe Tasks Vision for real-time oral-motor landmark tracking and articulatory movement verification during speech practice.
- **Reinforcement Learning Agent:** Developed a Q-learning adaptive difficulty controller dynamically tuning game environments based on user accuracy and reaction times.
- **High-Performance Architecture:** Delivered a modular FastAPI and WebSocket backend streaming low-latency perceptual feedback to React client applications.
- **Real-World Impact:** Shipped sub-50ms visual tracking biofeedback for pediatric therapy across vaaksudhi.manaslearning.com and vaakify.manaslearning.com.

Nivesh AI — AI/ML Engineer Intern

Jun 2026 – Present | Mumbai, India

- **Streaming Audio & Telemetry Processing:** Engineered scalable data pipelines on Azure handling streaming audio feeds and document ingestion for 10,000+ mutual fund distributors.
- **Spectral Signal Processing:** Implemented digital signal preprocessing (`noisereduce`) and acoustic filtering pipelines to isolate voice signals in high-noise environments.
- **Real-World Impact:** Automated unstructured audio data pipelines on cloud infrastructure, processing multi-hour client advisor calls with low latency.

FEATURED PROJECTS

NIRVANA (Firebot) — Autonomous Firefighting Robot & MuJoCo Simulation

[Live Render Console](#) · [GitHub: Firebot](#)

Python, MuJoCo 3D, PyTorch, Gymnasium, EIF Sensor Fusion, Lidar SLAM, FastAPI, WebSockets

- Developed an autonomous firefighting robot operating in a physics-accurate MuJoCo 3D simulator across procedural 8-room environments with volumetric particle fire modeling.
- Designed **MM-FusionRL** (multimodal cross-attention deep RL) coupled with an **Extended Information Filter (EIF)** for thermal localization, lidar motion planning, and autonomous obstacle navigation.
- Built a web mission console streaming real-time robot telemetry, lidar sweeps, and fire-event logs via WebSockets deployed live on Render.
- **Real-World Impact:** Eliminates hazardous human entry into structural fires by achieving autonomous 8-room thermal exploration and obstacle avoidance within 45 seconds in simulation.

Tripwire — Real-Time CPU-Only Computer Vision Vehicle Counter

[Live Demo: tripwire-inky.vercel.app](#) · [GitHub: Vehicle-Counter](#)

Python, OpenCV, FastAPI, React, Vite, WebSockets, Docker

- Engineered an edge-ready, GPU-free computer vision pipeline processing live CCTV and RTSP feeds with sub-50ms latency.
- Implemented Gaussian background subtraction, morphological noise filtering, centroid-based multi-object tracking, and directional virtual tripwire crossing counters streamed over WebSockets to a Vite console.
- **Real-World Impact:** Reduces smart city surveillance deployment costs by 80% by executing high-accuracy vehicle tracking on cheap commodity CPUs without requiring GPUs.

GECO-Enhanced — Confidence-Aware Semantic Keypoint Correspondence

[GitHub: GECO](#)

PyTorch, DINOv2 (ViT), Python Optimal Transport (POT), FastAPI, React, Docker

- Formulated an unsupervised semantic correspondence framework establishing keypoints across different instances of identical object classes under severe pose and perspective variation.
- Extracted dense DINOv2 ViT features and solved correspondence using Sinkhorn-regularized **Optimal Transport (POT)** with confidence weighting and entropy regularization.

- **Real-World Impact:** Enables zero-shot visual alignment across varying camera viewpoints without requiring expensive annotated paired training datasets.

Vaaksudhi — Pediatric Speech & Diagnostic Multi-Agent Platform

[Live Platform](#) · [GitHub: Agentic-AI](#)

Python, LangGraph, Faster-Whisper, Vosk, Kokoro ONNX, Librosa, MediaPipe, Q-Learning, FastAPI

- Architected an autonomous multi-agent clinical diagnostic pipeline coordinating 4 clinical assessment modules (BreathQuest, VaakMirror, Chime, VoiceHurdleRace) across 8 Indian languages.
- Implemented MediaPipe visual biofeedback and LangGraph state machines evaluating place, manner, and voicing errors with sub-300ms latency.
- **Real-World Impact:** Deployed clinical assessment system actively used for objective pediatric speech screening across 8 Indian languages.

Vaakify — Pediatric Multilingual Speech Practice Platform

[Live Platform](#) · [GitHub: Speech-Repeater](#)

Python, Librosa, Praat-Parselmouth, Faster-Whisper, Coqui TTS, FastAPI, React

- Engineered a phoneme-level articulation engine extracting jitter, shimmer, formant frequencies (F1/F2), and pitch contours from speech, powering clinical practice in English, Hindi, and Kannada.
- Integrated automated phonetic transcription via Epitran G2P to measure distance metrics against target speech and generate targeted phoneme drills.
- **Real-World Impact:** Provides objective, quantitative articulatory biofeedback directly to pediatric speech therapists, replacing subjective manual error scoring.

EDUCATION & CERTIFICATIONS

- **Bachelor of Technology (B.Tech)**, SASTRA University (Expected 2027)
- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems (*In-Person Practical Program*)
- **IIoT Certification:** Ministry of Heavy Industries (MHI) Program (*Project-Based Evaluation*)